

Arhitektura računalā

LV5

STATUS register

Bit	7	6	5	4	3	2	1	0
Flag	I	T	H	S	V	N	Z	C

- 8 bitni registar koji se nalazi u I/O dijelu memorije
- Sadrži 8 zastavica (flags) čija se vrijednost (0 ili 1) mijenja u ovisnosti o predhodnoj instrukciji

STATUS register

Bit	7	6	5	4	3	2	1	0
Flag	I	T	H	S	V	N	Z	C

Bit 0: Carry Flag

Bit 1: Zero Flag

Bit 2: Negative Flag

Bit 3: Two's Complement Overflow Flag

Bit 4: Sign Bit

Bit 5: Half Carry Flag

Bit 6: Bit Copy Storage

Bit 7: Global Interrupt Enable

STATUS register

CARRY FLAG (C)

- Postavlja se u slučaju preliva kod aritmetičke ili logičke operacije

Primjer

ldi r16,0xFF ; load r16 with 0xFF

ldi r17,0xFF ; load r17 with 0xFF

add r16,r17 ; carry will be set

Zbroj vrijednosti u registrima je veći od 8 bita → MSB se upisuje u Carry zastavicu.

STATUS register

CARRY FLAG (C)

Primjer ldi r16,0x01 ; load r16 with 0x01
 ldi r17,0x02 ; load r17 with 0x02
 add r16,r17 ; carry will not be set

Carry bit se ne aktivira

STATUS register

CARRY FLAG (C)

Primjer LSL

```
ldi r16,0b100000000 ; load 0b100000000 into r16  
lsl r16 ; carry will be set
```

```
ldi r16,0b000000000 ; load 0b000000000 into r16  
lsl r16 ; carry will not be set
```

STATUS register

ZERO FLAG (Z)

- Aktivira se ako je rezultat operacije jednak nula (0).

Primjer

ldi r16,0x03 ; load r16 with 0x03

dec r16 ; zero flag not set (r16 = 0x02)

dec r16 ; zero flag not set (r16 = 0x01)

dec r16 ; zero flag set (r16 = 0x00)

STATUS register

NEGATIVE FLAG (N)

- Aktivira se ako je 7 bit rezultata jednak 1.

Primjer

ldi r16,1 ; load r16 with 0x01

sbi r16,2 ; negative flag set (result = -1 or 255)

VAŽNO!

Podaci u memoriji su sekvence nula i jedinica. Koja vrsta podatka će to biti određuje programer načinom na koji će ga iskoristiti.

(NE signalizira je li podatak SIGN ili UNSIGN)

STATUS register

TWOS COMPLEMENT OVERFLOW FLAG (V)

- Aktivira se ako je rezultat SIGN podataka (-128 do 127) ne stane u 8 bita.

Primjer

```
ldi r16,-128 ; load r16 with -128  
dec r16 ; two's complement overflow set
```

```
ldi r17,127 ; load r17 with 127  
inc r17 ; two's complement overflow set
```

STATUS register

SIGN FLAG (S)

- Dodatni bit za utvrđivanje je li rezultat operacije pozitivan ili negativan

Primjer 1.

oduzimanje 10 od -120 (očekivani rezultat je -130 ali iznosi +126)

ldi r16,0x88 ; load r16 with 0x88 (-120)

ldi r17,0x0A ; load r17 with 0x0A (10)

sub r16,r17 ; subtract r17 from r16 (result = 0x7E = 126)

STATUS register

SIGN FLAG (S)

Primjer 2.

Zbrajanje 10 i 120 (očekivani rezultat je 130)

ldi r16,0x78 ; load r16 with 0x88 (120)

ldi r17,0x0A ; load r17 with 0x0A (10)

add r16,r17 ; add r17 to r16 (result = 0x82 = 130)

Rezultat je ispravan iznosi 130, ali je aktiviran Negative flag koji bi značio da je rezultat negativan. SIGN flag sad nije aktivan što znači da je rezultat pozitivan.

STATUS register

HALF CARRY FLAG (H)

- Aktivira se kad dolazi do prijenosa s donja 4 bita na gornja 4 bita registra

Primjer

```
ldi r16,0x0F ; load r16 with 0x0F  
inc r16
```

Koristan kod Binary Coded Digits kad su dva broja spremljena u jednom registru (jedan u donjem drugi u donjem dijelu registra).

STATUS register

COPY STORAGE FLAG (T)

- Ne postavlja se nijednom aritmetičko ili logičkom operacijom već pomoću SET i CLT instrukcija

Primjer

set ; set T flag

clt ; clear T flag

Koristan kod korištenja subrutina. Je li rutina uspješno izvršena ili ne...

STATUS register

GLOBAL INTERRUPT ENABLE FLAG (I)

- Koristi se za upravljanje prekidima (interrupt). Ako je bit aktivan (1) omogućeni su prekidi u suprotni su onemogućeni.

Primjer

```
sei ; enable interrupts  
cli ; disable interrupts
```

JMP

JMP – Jump

Description

Jump to an address within the entire 4M (words) Program memory. See also [RJMP](#).

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $PC \leftarrow k$

Syntax:

Operands:

Program Counter:

Stack:

(i) JMP k

$0 \leq k < 4M$

$PC \leftarrow k$

Unchanged

32-bit Opcode:

1001	010k	kkkk	110k
kkkk	kkkk	kkkk	kkkk

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

RJMP

RJMP – Relative Jump

Description

Relative jump to an address within $PC - 2K + 1$ and $PC + 2K$ (words). For AVR microcontrollers with Program memory not exceeding 4K words (8KB) this instruction can address the entire memory from every address location. See also JMP.

Operation:

(i) $PC \leftarrow PC + k + 1$

Syntax:

Operands:

Program Counter:

Stack:

(i) RJMP $k - 2K \leq k < 2K$ $PC \leftarrow PC + k + 1$ Unchanged

16-bit Opcode:

1100	kkkk	kkkk	kkkk
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Zadatak 1

- Realiziraj program koji najprije povećava registar R1 zatim skače na adresu 0x0040 (potrebna organizacija memorije!) u programskoj memoriji te izvršava 5 NOP instrukcija. Nakon izvršenja vraća se na početak programa.

Rješenje zadatka 1.

start:

inc r16

rjmp 0x0040

.org 0x0040

NOP

NOP

NOP

NOP

NOP

jmp start

IJMP

IJMP – Indirect Jump

Description

Indirect jump to the address pointed to by the Z (16 bits) Pointer Register in the Register File. The Z-pointer Register is 16 bits wide and allows jump within the lowest 64K words (128KB) section of Program memory.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:		Comment:		
(i)	$PC \leftarrow Z(15:0)$	Devices with 16-bit PC, 128KB Program memory maximum.		
(ii)	$PC(15:0) \leftarrow Z(15:0)$ $PC(21:16) \leftarrow 0$	Devices with 22-bit PC, 8MB Program memory maximum.		
Syntax:		Operands:	Program Counter:	Stack:
(i), (ii) IJMP		None	See Operation	Not Affected

16-bit Opcode:

1001	0100	0000	1001
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Zadatak 2. (isti zadatak)

- Realiziraj program koji najprije povećava registar R1 zatim skače na adresu 0x0040 (potrebna organizacija memorije!) u programskoj memoriji te izvršava 5 NOP instrukcija. Nakon izvršenja vraća se na početak programa.
- Dodatna uputa
 - Potrebno u registar Z pohraniti adresu na koju skaćemo!

Rješenje - zadatak 2.

start:

inc r16

ldi ZL, 0x40

ldi ZH, 0x00

ijmp

.org 0x00400

NOP

NOP

NOP

NOP

NOP

jmp start

BRNE

BRNE – Branch if Not Equal

Description

Conditional relative branch. Tests the Zero Flag (Z) and branches relatively to PC if Z is cleared. If the instruction is executed immediately after any of the instructions CP, CPI, SUB, or SUBI, the branch will occur if and only if, the unsigned or signed binary number represented in Rd was not equal to the unsigned or signed binary number represented in Rr. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBC 1,k.)

Operation:

- (i) If $Rd \neq Rr$ ($Z = 0$) then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

- (i) BRNE k

Operands:

$$-64 \leq k \leq +6$$

Program Counter:

$$PC \leftarrow PC + k + 1$$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	01kk	kkkk	k001
------	------	------	------

Primjer BRNE

start:

LDI r16, 0b00000000

LDI r17, 0b00000000 ; ili 0b00000001

add R16,R17

BRNE nastavak

NOP

NOP

.org 0x0040

nastavak:

NOP

NOP

NOP

NOP

NOP

jmp start

BREQ – Branch if Equal

BREQ

- Suprotno od BRNE

Description

Conditional relative branch. Tests the Zero Flag (Z) and branches relatively to PC if Z is set. If the instruction is executed immediately after any of the instructions CP, CPI, SUB, or SUBI, the branch will occur if and only if the unsigned or signed binary number represented in Rd was equal to the unsigned or signed binary number represented in Rr. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBS 1,k.)

Operation:

- (i) If $R_d = R_r$ ($Z = 1$) then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

- (i) BREQ k

Operands:

$$-64 \leq k \leq +63$$

Program Counter:

$$PC \leftarrow PC + k + 1$$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	00kk	kkkk	k001
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Primjer BREQ

start:

`LDI r16, 0b00000000`

`LDI r17, 0b00000000 ; ili 0b00000001`

`add R16,R17`

`BREQ nastavak`

`NOP`

`NOP`

`.org 0x0040`

nastavak:

`NOP`

`NOP`

`NOP`

`NOP`

`NOP`

`jmp start`

BRSH

BRSH – Branch if Same or Higher (Unsigned)

Description

Conditional relative branch. Tests the Carry Flag (C) and branches relatively to PC if C is cleared. If the instruction is executed immediately after execution of any of the instructions CP, CPI, SUB, or SUBI, the branch will occur if and only if, the unsigned binary number represented in Rd was greater than or equal to the unsigned binary number represented in Rr. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBC 0,k.)

Operation:

- (i) If $R_d \geq R_r$ ($C = 0$) then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

- (i) BRSH k

Operands:

$$-64 \leq k \leq +63$$

Program Counter:

$$PC \leftarrow PC + k + 1$$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	01kk	kkkk	k000
------	------	------	------

BRLO

BRLO – Branch if Lower (Unsigned)

Description

Conditional relative branch. Tests the Carry Flag (C) and branches relatively to PC if C is set. If the instruction is executed immediately after any of the instructions CP, CPI, SUB, or SUBI, the branch will occur if and only if, the unsigned binary number represented in Rd was smaller than the unsigned binary number represented in Rr. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBS 0,k.)

Operation:

- (i) If $R_d < R_r$ ($C = 1$) then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

- (i) BRLO k

Operands:

$$-64 \leq k \leq +63$$

Program Counter:

$$PC \leftarrow PC + k + 1$$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	00kk	kkkk	k000
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

BRLT

BRLT – Branch if Less Than (Signed)

Description

Conditional relative branch. Tests the Signed Flag (S) and branches relatively to PC if S is set. If the instruction is executed immediately after any of the instructions CP, CPI, SUB, or SUBI, the branch will occur if and only if, the signed binary number represented in Rd was less than the signed binary number represented in Rr. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBS 4,k.)

Operation:

- (i) If $R_d < R_r$ ($N \oplus V = 1$) then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

Operands:

Program Counter:

- (i) BRLT k

$-64 \leq k \leq +63$

$PC \leftarrow PC + k + 1$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	00kk	kkkk	k100
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

BRMI

BRMI – Branch if Minus

Description

Conditional relative branch. Tests the Negative Flag (N) and branches relatively to PC if N is set. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form. (Equivalent to instruction BRBS 2,k.)

Operation:

- (i) If $N = 1$ then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

- (i) BRMI k

Operands:

$$-64 \leq k \leq +63$$

Program Counter:

$$PC \leftarrow PC + k + 1$$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	00kk	kkkk	k010
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

BRBC

BRBC – Branch if Bit in SREG is Cleared

Description

Conditional relative branch. Tests a single bit in SREG and branches relatively to PC if the bit is cleared. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form.

Operation:

- (i) If $SREG(s) = 0$ then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

Operands:

Program Counter:

- (i) BRBC s, k

$0 \leq s \leq 7, -64 \leq k \leq +63$

$PC \leftarrow PC + k + 1$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	01kk	kkkk	ksss
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

BRBS – Branch if Bit in SREG is Set

BRBS

Description

Conditional relative branch. Tests a single bit in SREG and branches relatively to PC if the bit is set. This instruction branches relatively to PC in either direction ($PC - 63 \leq \text{destination} \leq PC + 64$). Parameter k is the offset from PC and is represented in two's complement form.

Operation:

- (i) If $SREG(s) = 1$ then $PC \leftarrow PC + k + 1$, else $PC \leftarrow PC + 1$

Syntax:

Operands:

Program Counter:

- (i) BRBS s,k

$0 \leq s \leq 7, -64 \leq k \leq +63$

$PC \leftarrow PC + k + 1$

$PC \leftarrow PC + 1$, if condition is false

16-bit Opcode:

1111	00kk	kkkk	ksss
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Instrukcije grananja (ovisno o jednom bitu SREG)

- **BRCC – Branch if Carry Cleared**
- **BRCS – Branch if Carry Set**
- **BRHC – Branch if Half Carry Flag is Cleared**
- **BRHS – Branch if Half Carry Flag is Set**
- **BRID – Branch if Global Interrupt is Disabled**
- **BRIE – Branch if Global Interrupt is Enabled**
- **BRPL – Branch if Plus**
- **BRMI – Branch if Minus**
- **BRTC – Branch if the T Flag is Cleared**
- **BRTS – Branch if the T Flag is Set**
- **BRVC – Branch if Overflow Cleared**
- **BRVS – Branch if Overflow Set**